ARAVINDAN RAMESHBABU

SENIOR DATA & AI ENGINEER

+91-9659959575 • aravindanramesh910@gmail.com • linkedin.com/in/aravindanramesh • github.com/arv1ndofficial • Chennai, India

PROFESSIONAL SUMMARY

Senior Data & AI Engineer with **5+ years** of expertise designing scalable cloud data platforms, distributed lakehouses, and applied GenAI workflows. Technical lead for a 5-engineer team operating Apache Iceberg and Spark on AWS EKS; specialist in multi-tenant orchestration, SCD Type 1 & 2 history pipelines, Databricks Auto Loader, Delta Live Tables (DLT), and Model Context Protocol (MCP) integrations for autonomous data management.

TECHNICAL SKILLS

Big Data & Lakehouse: Apache Spark (PySpark, Spark SQL, Structured Streaming), Apache Iceberg, Delta Lake, Trino, Nessie Catalog, Hive Metastore

Streaming & Ingestion: Apache Kafka, Spark Streaming, Databricks Auto Loader, Change Data Feed (CDF), Declarative CDC (SCD 1 & 2)

Cloud & Infrastructure: AWS (EKS, S3, Glue, Lambda, EC2, IAM / IRSA), Databricks, GCP, Kubernetes, Docker, GitHub Actions, Linux / Bash

AI & GenAI Platforms: Model Context Protocol (MCP), RAG Architectures, ChromaDB, Ollama, Open-Source LLMs, SeqGAN Synthetic Generation

Databases & Query: PostgreSQL, MySQL, MongoDB, Cassandra, Dremio, SQL

Languages & Architecture: Python, SQL, PySpark, Scala, Multi-Tenant Architecture, Medallion Architecture, CI/CD

WORK EXPERIENCE

Senior Data Engineer (SDE 2) · Engineering Team Lead

Feb 2025 – Present

M2P Fintech · Chennai, India

- **Open-Source Lakehouse:** Architected and deployed an AWS EKS lakehouse with Spark, Iceberg, Trino, and Nessie Catalog, cutting query latency and cloud storage overhead.
- **Autonomous MCP Agents:** Implemented Model Context Protocol (MCP) tooling, empowering LLM agents to trigger, query, and monitor ETL pipelines via natural language.
- **Fintech SCD-2 Compliance:** Engineered SCD Type 2 history pipelines on Iceberg with dynamic watermarking and Hot/Cold partitioning for zero-loss audit compliance.
- **Serverless Ingestion:** Automated AWS Glue and Lambda ingestion pipelines with schema validation, reducing manual schema maintenance by **40%**.
- **Security & Governance:** Enforced IRSA fine-grained access controls for Spark and Trino on EKS; configured webhook alerting for immediate failure remediation.
- **Leadership & Recognition:** Led a 5-engineer team delivering the lakehouse ahead of schedule; earned the **Above & Beyond in Engineering (Applause 2026)** award.

Data Engineer

Apr 2021 – Feb 2025

Tata Consultancy Services · Chennai, India

- **Kubernetes Spark Optimization:** Developed and tuned large-scale PySpark data processing workloads on Kubernetes Spark clusters for telecom analytics.
- **Workflow Orchestration:** Automated resilient ETL pipelines using Apache Airflow DAGs on Kubernetes, improving cluster resource utilization.
- **Cloud Migration:** Spearheaded migration of legacy Oracle databases to GCP Cloud using PySpark and Spark SQL, accelerating job runtimes.
- **Storage & Query Tuning:** Designed partitioned, compressed Hive tables across AWS S3 and GCP, reducing storage footprint and read latency.
- **Real-Time Kafka Streaming:** Built Spark Streaming pipelines for high-throughput telecom event consumption; integrated Dremio for federated querying.
- **Awards:** Honored with **Technical Excellence (Aug 2022)** and **Inspiring Talent (Jun 2023)** for outstanding project delivery and innovation.

PROFESSIONAL PROJECTS

Prism & Nexus ETL — Agentic Control Plane & Multi-Tenant Lakehouse Feb 2025 – Present

M2P Fintech · Spark, Iceberg, Trino, Airflow, Unity Catalog, Model Context Protocol (MCP)

- **Platform Architecture:** Built unified control plane hosting Airflow, Spark, Trino, and Unity Catalog orchestrating multi-source ingestion into Iceberg and Delta formats.
- **Natural Language MCP Tooling:** Exposed connections, migrations, and pipeline execution as MCP tools for natural language agentic data operations.
- **Schema Evolution:** Developed multi-tenant ingestion with staged primary-key upserts, dynamic drift detection, and duplicate suppression.

SCD Type 2 History & Hot/Cold Table Architecture Feb 2025 – Present

M2P Fintech · Apache Iceberg, Dynamic Watermarking, Hive Metastore, Spark SQL

- **Hot/Cold Table Design:** Engineered Hot Table for rolling-window upserts and automated Cold Aggregates ETL for KPI roll-ups over impacted date partitions.
- **Audit-Grade Compliance:** Established automated namespace lifecycle management on Iceberg with dynamic watermarking for immutable fintech audit trails.

Synthetic Telecom Data & RAG / MCP Orchestration Platform Jan 2023 – Feb 2025

Tata Consultancy Services · SeqGAN, RAG Architectures, ChromaDB, Ollama, Apache Airflow, Python

- **Generative Synthetic Data:** Developed SeqGAN deep learning architecture generating high-fidelity synthetic telecom data to eliminate training bias.
- **Enterprise RAG Retrieval:** Built semantic retrieval using ChromaDB and Ollama LLMs, orchestrated via Airflow with MCP agent bindings.

Real-Time Location-Based Marketing Stream Apr 2021 – Dec 2022

Tata Consultancy Services · Spark Streaming, Apache Kafka, Cassandra NoSQL, GDPR Compliance

- **Real-Time Spatial Stream:** Ingested high-velocity ISP tower location events with Spark Streaming to generate sub-second geo-targeted promotions.
- **GDPR Audited Storage:** Designed Cassandra NoSQL sliding-window store with automated 6-month TTL cleanup, delivering compliant real-time partner feeds.

PERSONAL PROJECTS

Goodcabs — Transportation Lakehouse & Declarative CDC Pipeline 2025

Databricks · Spark Declarative Pipelines (DLT) · Auto Loader · Delta Lake · SCD Type 1 CDC

- **Medallion Lakehouse on Databricks:** Architected Bronze-Silver-Gold pipeline using Spark Declarative Pipelines processing cab transactions across 10 cities.
- **Auto Loader Streaming Ingestion:** Deployed Auto Loader (`cloudFiles`) with automated schema inference and rescue drift handling.
- **Declarative CDC Flow:** Engineered SCD Type 1 CDC upserts via `dp.create_auto_cdc_flow` with `sequence_by` ordering to resolve out-of-order records.
- **Quality Gates & Dimensions:** Enforced data quality expectations (`@dp.expect`), dynamic calendar dimension generation, and pre-filtered Gold regional marts.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics & Communication 2016 – 2020

University College of Engineering Tindivanam, Anna University · Tamil Nadu, India

- Graduated First Class (CGPA 7.05); final-year OpenCV Facial Recognition project awarded 'S' (Top Grade).
- Elected Student Cell Chairman; directed university-wide technical symposiums.

CERTIFICATIONS & CREDENTIALS

- **Data Streaming Engineer — Apache Flink** – Confluent (Apr 2026 – Apr 2028)
- **Azure Data Engineer Associate (DP-203)** – Microsoft
- **AWS Solutions Architect — Associate** – Amazon Web Services
- **Linux Essentials (LE-01)** – Linux Professional Institute
- **AI for Real-World Applications Specialization**

HONORS & AWARDS

- **Above & Beyond in Engineering (Applause 2026)** – M2P Fintech (Aug 2026)
- **Inspiring Talent Award** – Tata Consultancy Services (Jun 2023)
- **Technical Excellence Award** – Tata Consultancy Services (Aug 2022)